树莓派查看imu数据

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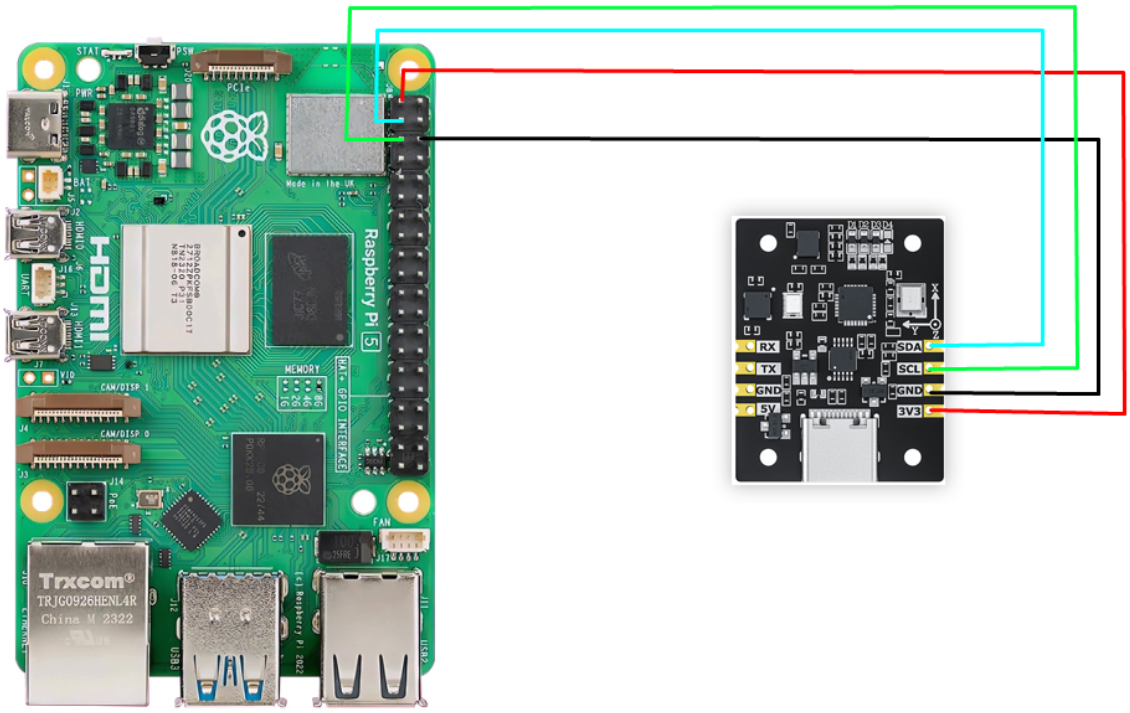
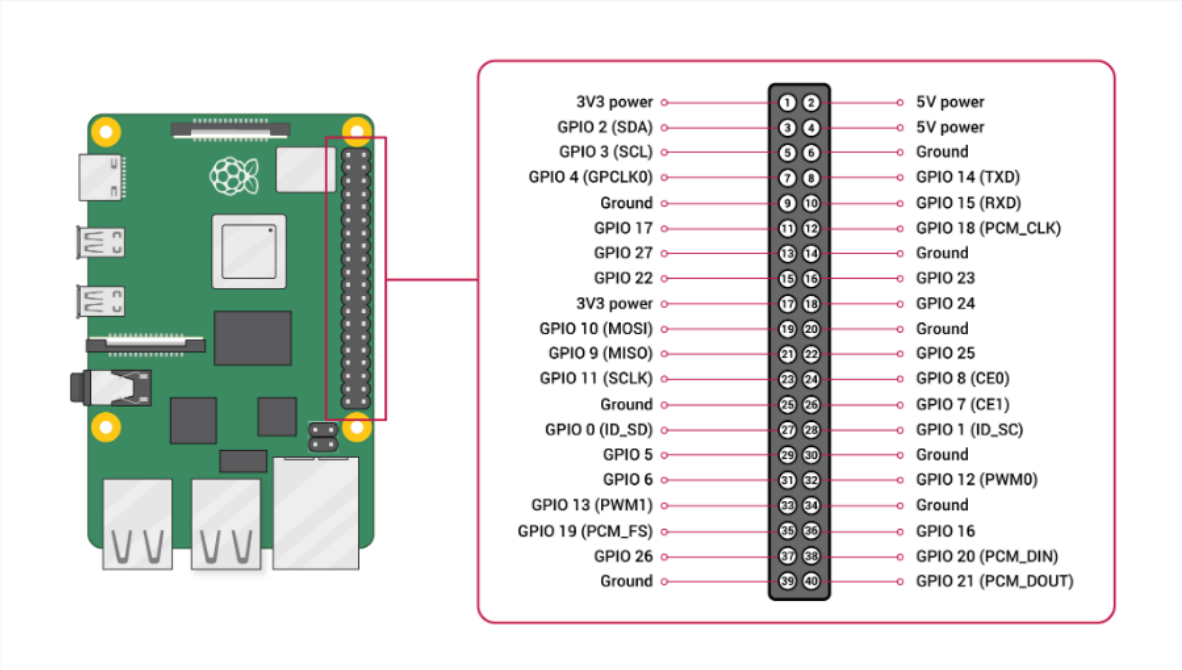
1.连接设备

本教程以树莓派5主板，官方64位版本的镜像为例。

将IMU姿态传感器按照下图连接到树莓派5的I2C接口。



IMU姿态传感器	树莓派5（物理引脚）
SDA	3
SCL	5
GND	GND
3.3V	3.3V



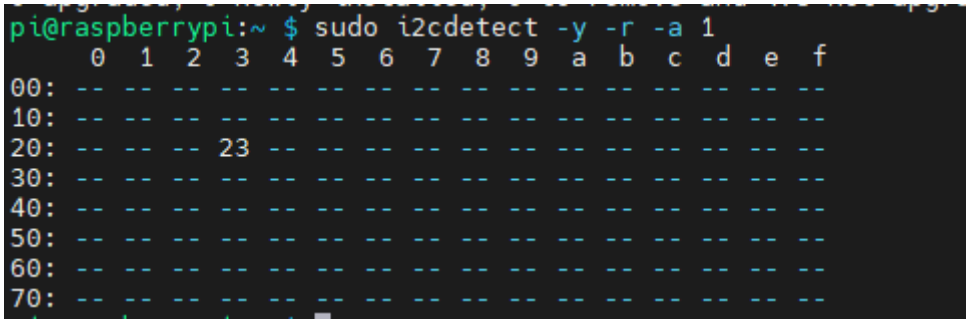
2.查看设备状态

首先安装 I2Ctool，终端输入：

```
sudo apt-get update
sudo apt-get install -y i2c-tools
```

查看I2C设备

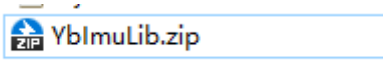
```
sudo i2cdetect -y -r -a 1
```



3.安装驱动库

3.1下载python驱动库文件

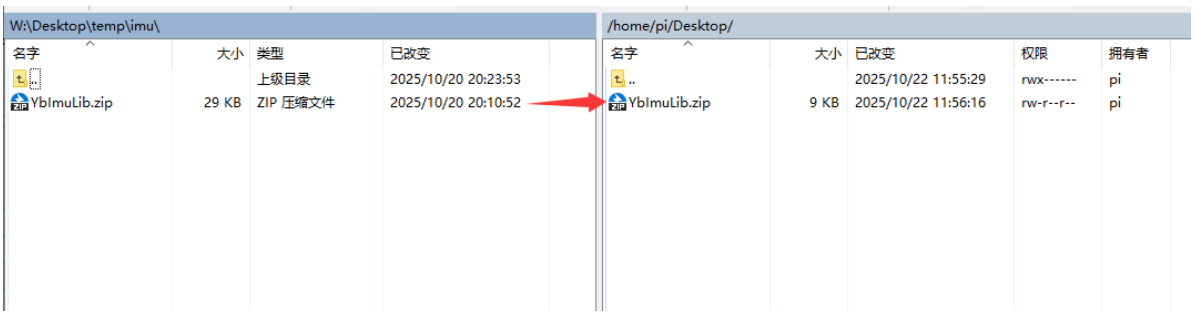
资料文件夹有提供最新版本的驱动库，名字为YbImuLib.zip



3.2传输文件

通过WinSCP软件将驱动库压缩包文件拖入Jetson桌面上。

安装成功后驱动库文件可删除掉。



如果还不会使用WinSCP传输文件的朋友，请查看以下网页WinSCP详细安装和操作方法：

文件传输

3.3安装驱动库

打开Jetson Nano的终端，输入以下命令解压。

进入桌面并查看文件是否存在，红色方框中就是目标文件

```
cd ~/Desktop && ls
```

```
pi@raspberrypi:~ $ cd ~/Desktop && ls
YbImuLib.zip
pi@raspberrypi:~/Desktop $
```

解压文件

```
unzip YbImuLib.zip
```

```
pi@raspberrypi:~/Desktop $ unzip YbImuLib.zip
Archive:  YbImuLib.zip
   creating: YbImuLib/
  inflating: YbImuLib/.gitignore
  inflating: YbImuLib/README.md
  inflating: YbImuLib/setup.py
   creating: YbImuLib/YbImuLib/
  inflating: YbImuLib/YbImuLib/__init__.py
  inflating: YbImuLib/YbImuLib/YbImuI2cLib.py
  inflating: YbImuLib/YbImuLib/YbImuSerialLib.py
pi@raspberrypi:~/Desktop $
```

进入驱动库文件夹

```
cd YbImuLib
```

运行安装命令，看到最后提示的安装版本号则表示安装成功。此命令会覆盖掉之前已经安装过的Rosmaster_Lib驱动库。

```
sudo python3 setup.py install
```

```
pi@raspberrypi:~/Desktop/YbImuLib $ sudo python3 setup.py install
running install
```

安装需要的库

```
sudo pip3 install pyserial
sudo pip3 install smbus2
```

4.查看imu数据

参考**3.2传输文件**的操作，用winscp将**YbImu_ReadData_I2C.py**文件传输到树莓派5上，然后运行命令

```
python3 YbImu_ReadData_I2C.py
```

```
pi@raspberrypi:~/Desktop $ python3 YbImu_ReadData_I2C.py
Firmware version: 0.0.9
Press Ctrl+C to exit the program.
------ Sensor Data ------
Acceleration [g]:       x= 0.000, y= 0.931, z=-0.256
Gyroscope [rad/s]:      x= 0.000, y= 0.000, z= 0.000
Magnetometer [uT]:      x=-31.251, y=-19.483, z= 48.390
Quaternion:             w=-0.28648, x=-0.38907, y=-0.69087, z=-0.53468
Euler Angle [deg]:      roll= 104.98, pitch=-1.16, yaw= 121.95
Barometer:              height= 0.00 m, temperature= 0.00 °C
                        pressure= 0.00000 Pa, pressure_diff= 0.00000 Pa
-------------------------
```

注意：以上为9轴IMU的数据读取，6轴无磁力计（Magnetometer）与气压计（Barometer）数据，9轴无气压计（Barometer）数据。

5.注意事项

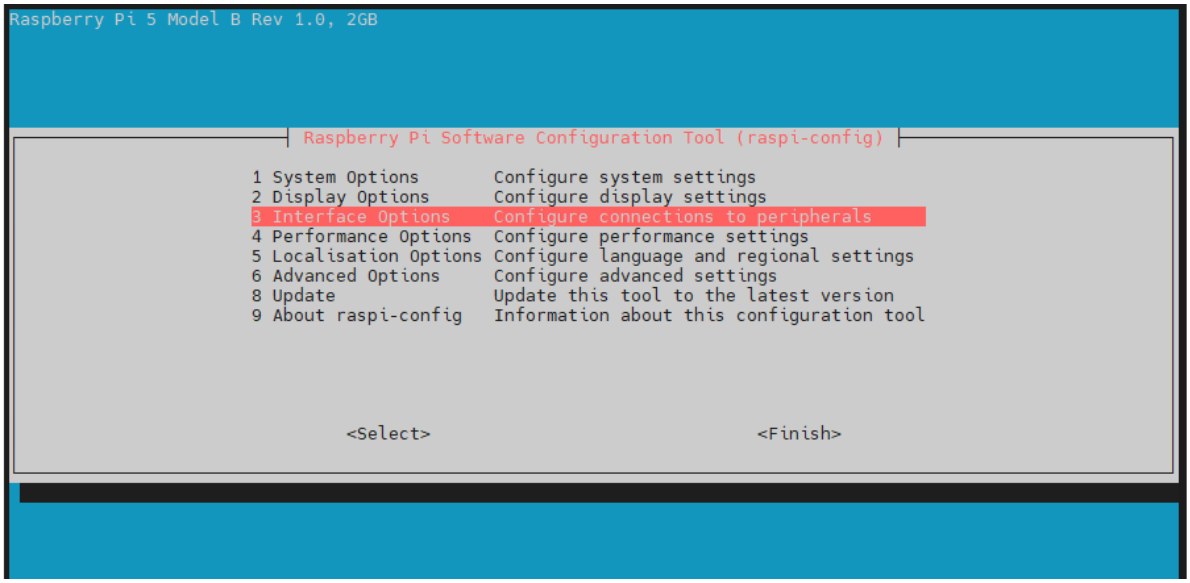
树莓派5需要提前开启i2c引脚。

开启操作如下：

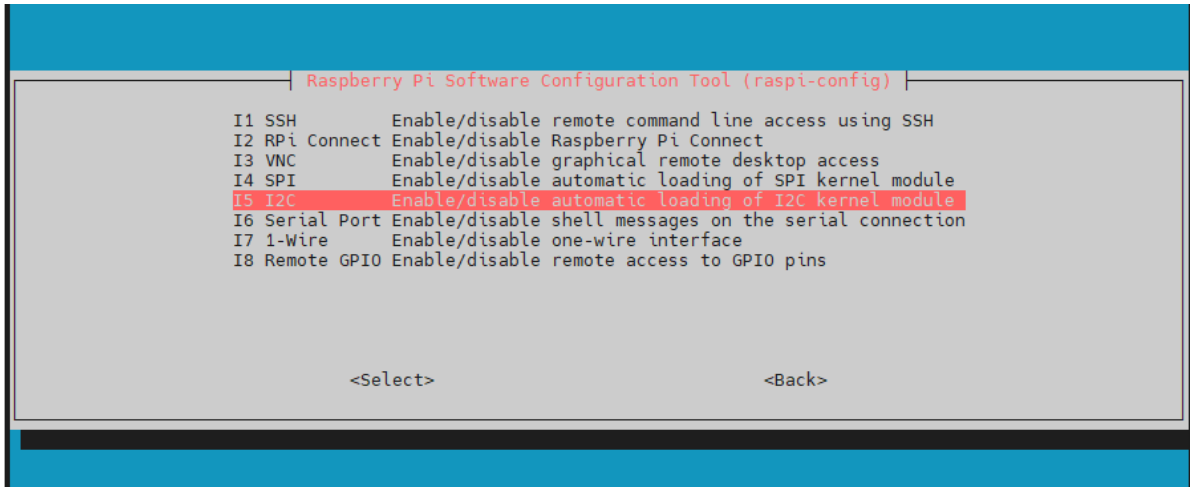
终端运行命令

```
sudo raspi-config
```

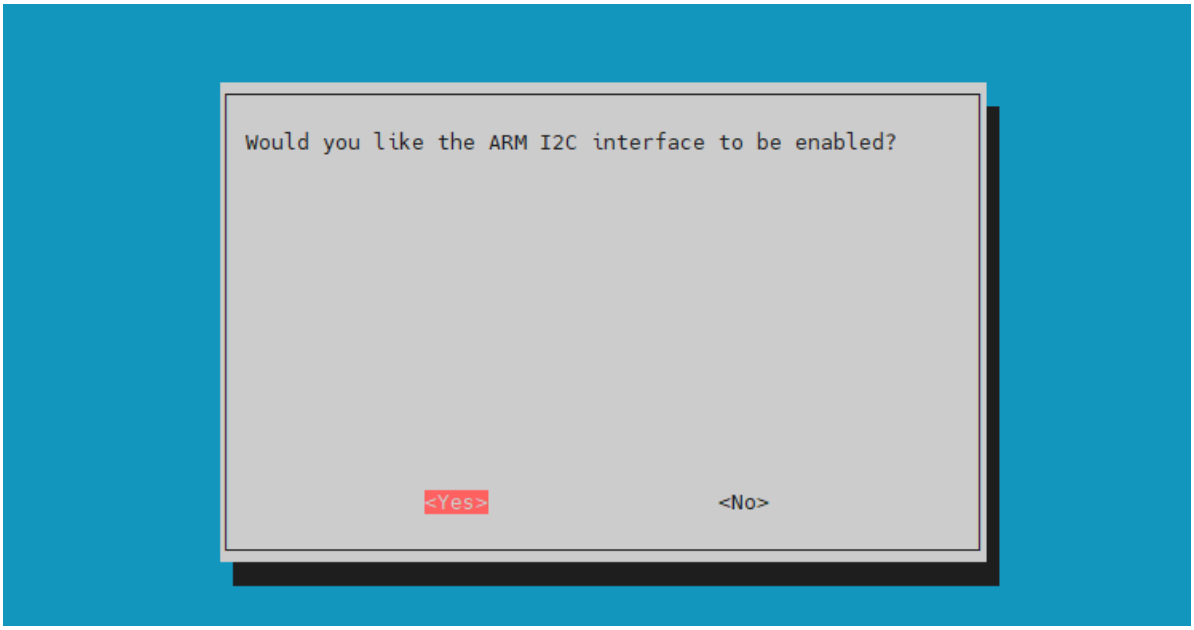
通过键盘的方向键选择，选中后按键盘Enter键进入



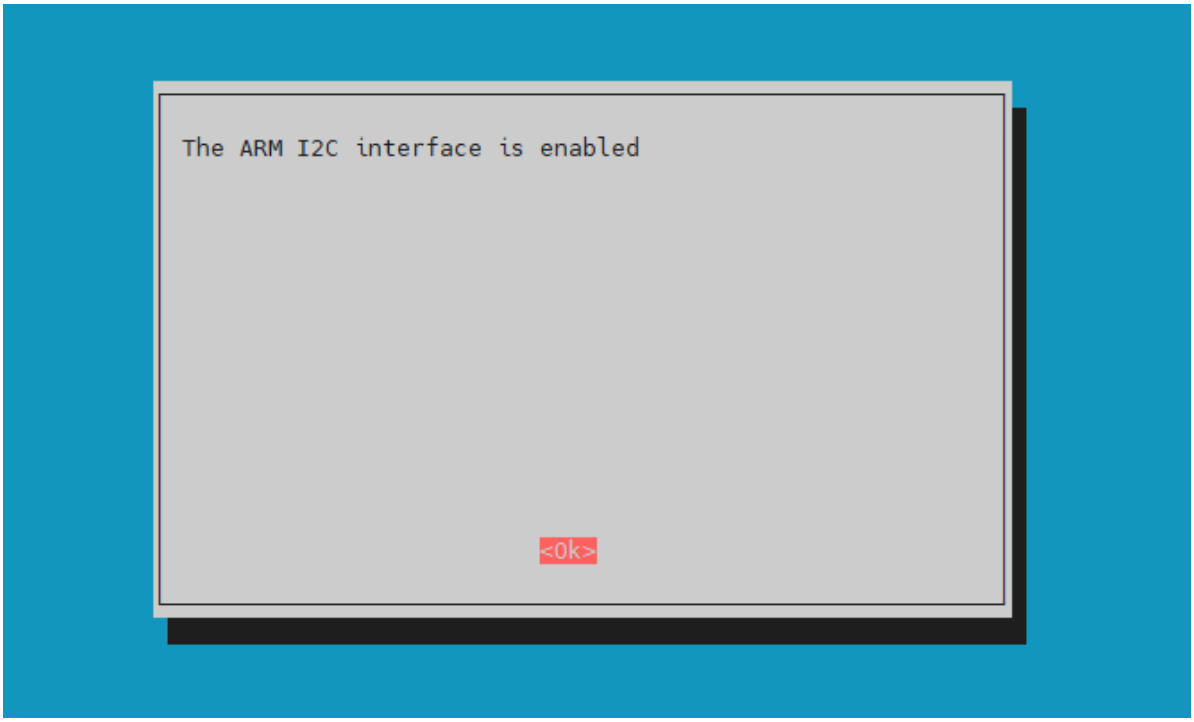
选择I2C，选中后按键盘Enter键进入，



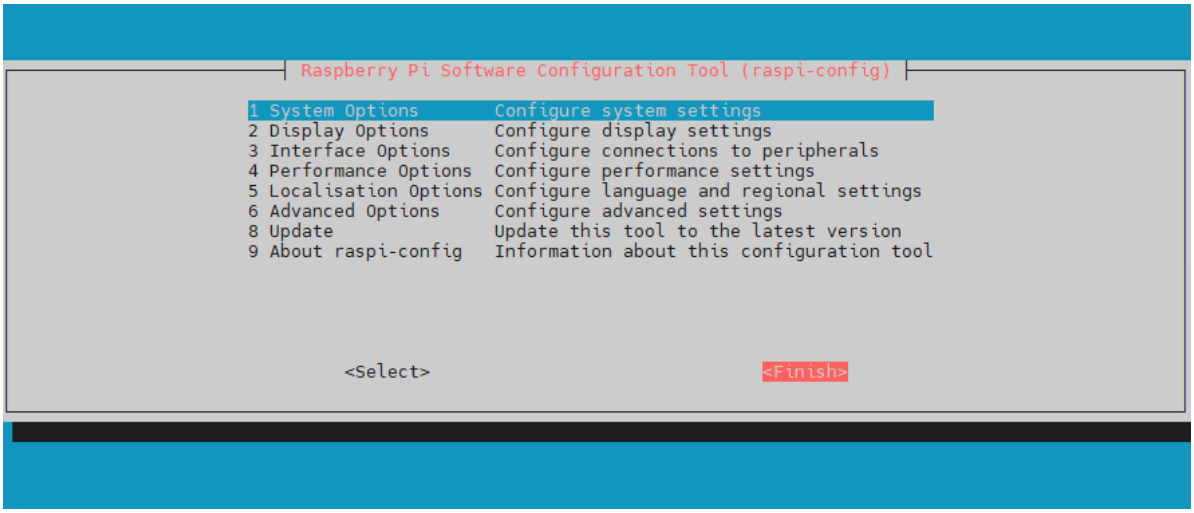
选中I2C后，按键盘Enter键，控制方向键选择Yes，再按Eneter键确认。



按Enter键确认



按方向键选择Finish，然后按Enter键，退出配置。


Raspberry Pi Software Configuration Tool (raspi-config)
1 System Options Configure system settings
2 Display Options Configure display settings
3 Interface Options Configure connections to peripherals
4 Performance Options Configure performance settings
5 Localisation Options Configure language and regional settings
6 Advanced Options Configure advanced settings
8 Update Update this tool to the latest version
9 About raspi-config Information about this configuration tool
<Select>
<Finish>